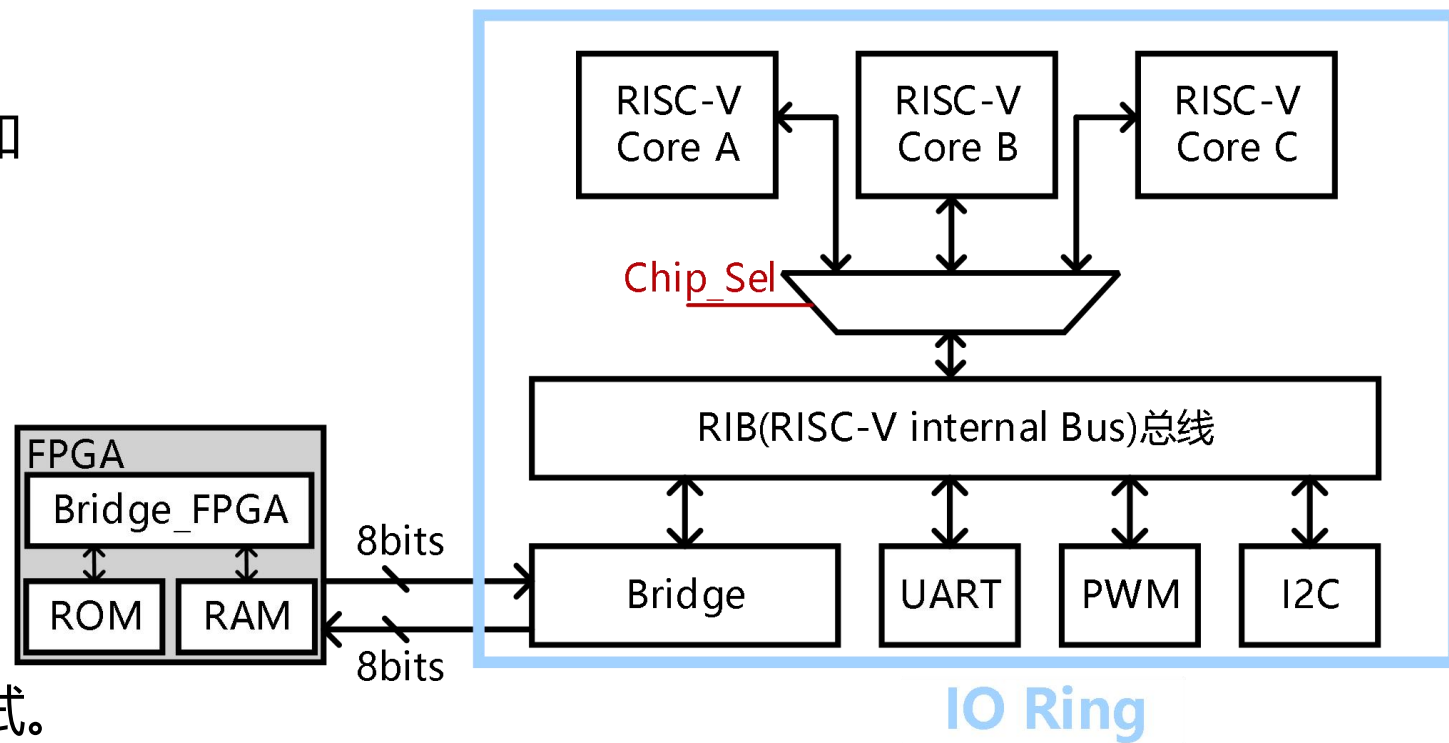


大作业题目：RISC-V处理器设计与实践

本作业基于开源RISC-V项目：tinyriscv (<https://gitee.com/liangkangnan/tinyriscv/tree/master>)

具体要求如下：

1. 每人独立的对tinyriscv进行修改，添加课程要求的拓展指令
2. 在FPGA上进行功能验证（助教提供用于运行指令）
3. 进行综合、仿真、后端等工作，提供DRC与LVS clean的gds文件
4. 对流片回来的芯片进行功能与性能测试。



***在前端设计阶段，两人使用一个FPGA，各自完成自己的设计。**

***为了提高面积的利用率，流片前需将三颗芯片放在一个IO Ring内部共用IO，通过chip_sel信号选择工作的芯片。**

因此三人一组，进行前端设计，一起进行布局布线，流片回来后的测试分开进行。

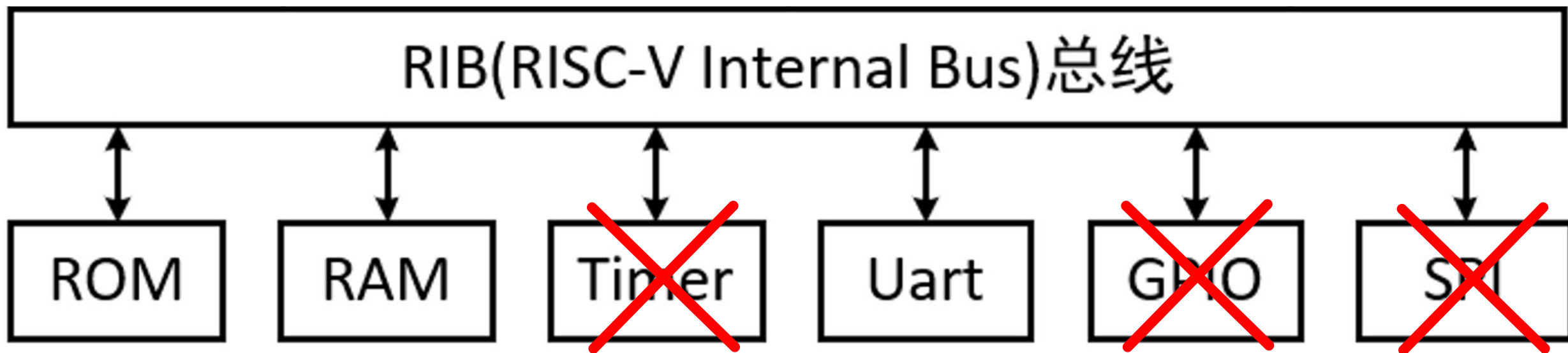
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- Tinyriscv 修改
- 时间节点
- FPGA 功能验证
- Vivado 下载
- FPGA 基本使用

Tinyriscv修改0-资源的删减

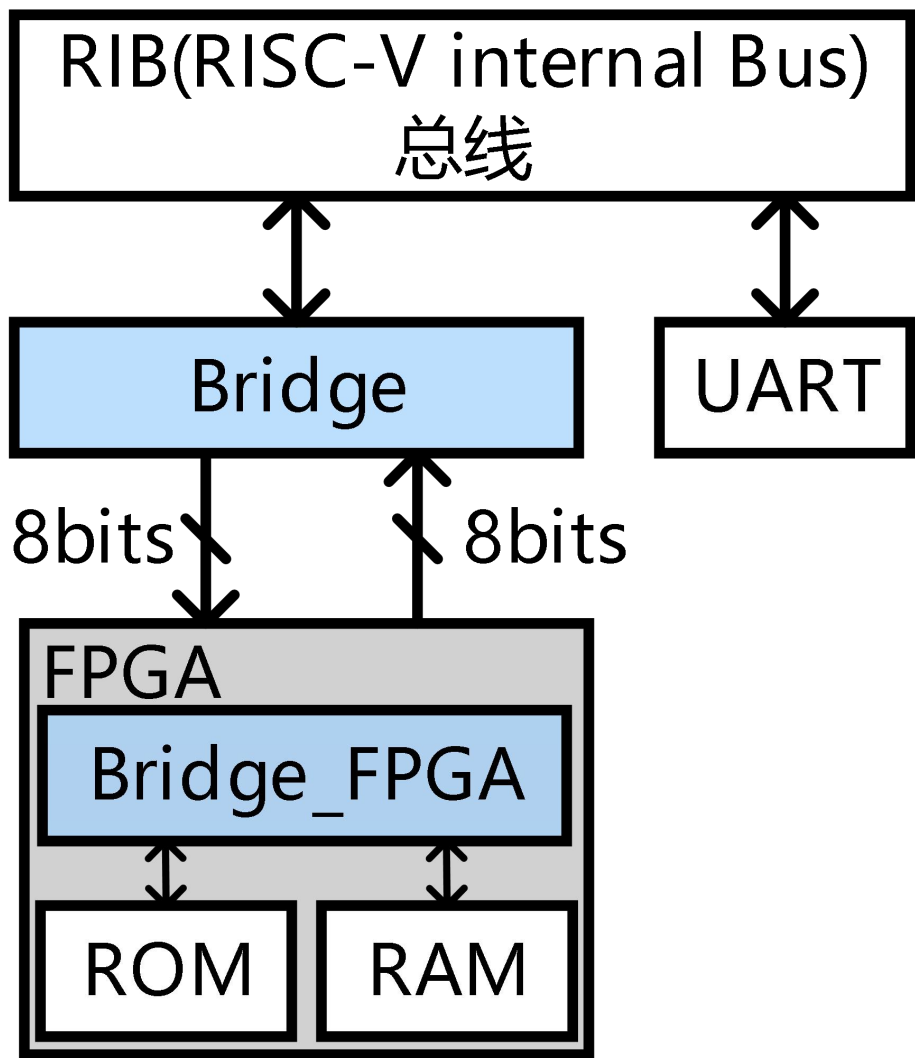
由于流片面积的限制，现需要对原有的开源代码进行以下修改

- 原有的 Tinyriscv中包含了一些外设，需要删除其中的Timer、SPI、GPIO外设
- 原有的 uart_debug 接收的 packet 大小为131，模块内部存在 132d x 8w 的寄存器组，现需修改为 uart_debug 接收的 packet 大小为35，模块内部存在 35d x 8w 的寄存器组



Tinyriscv修改1-ROM与RAM移到片外

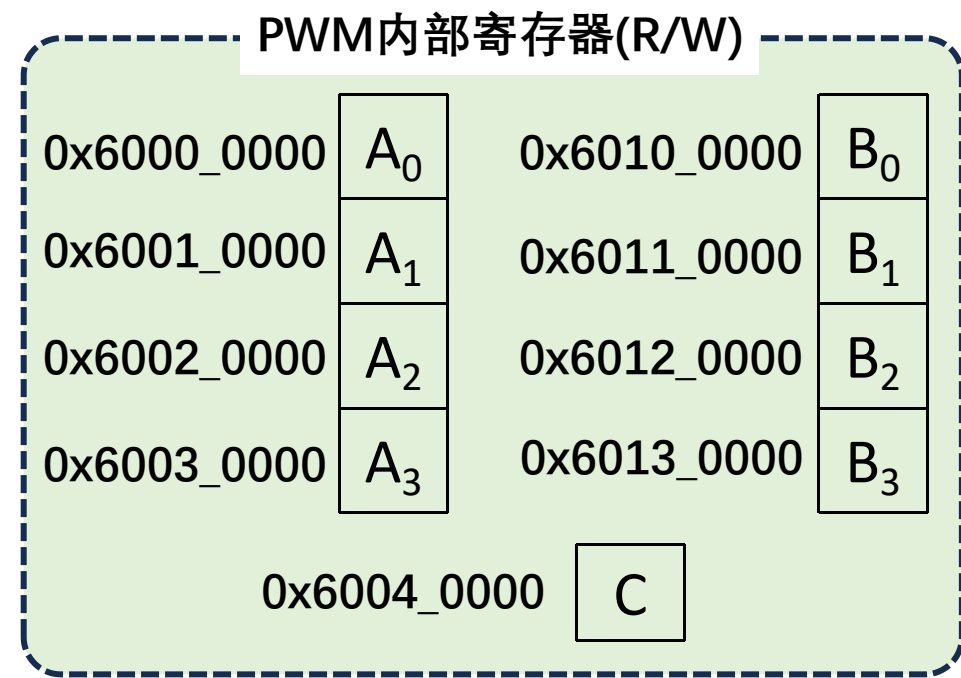
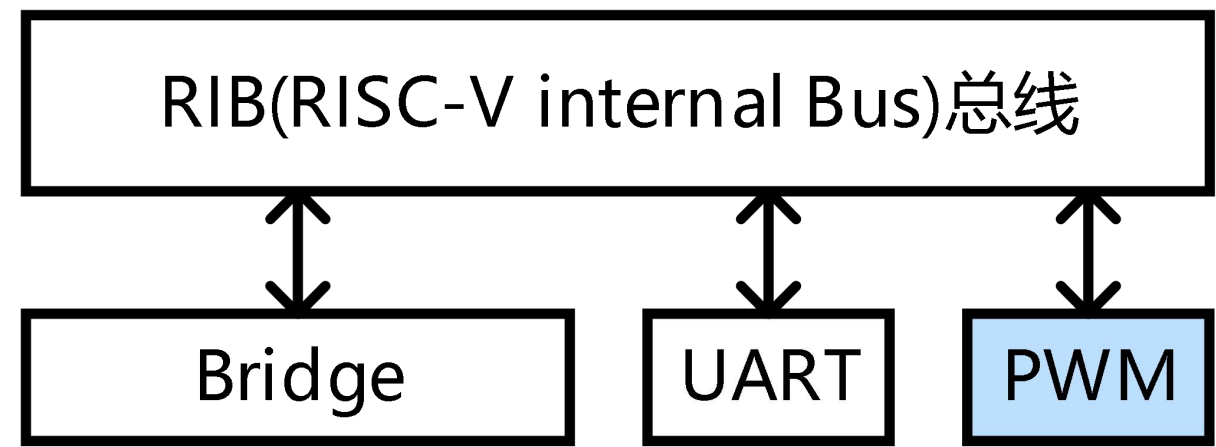
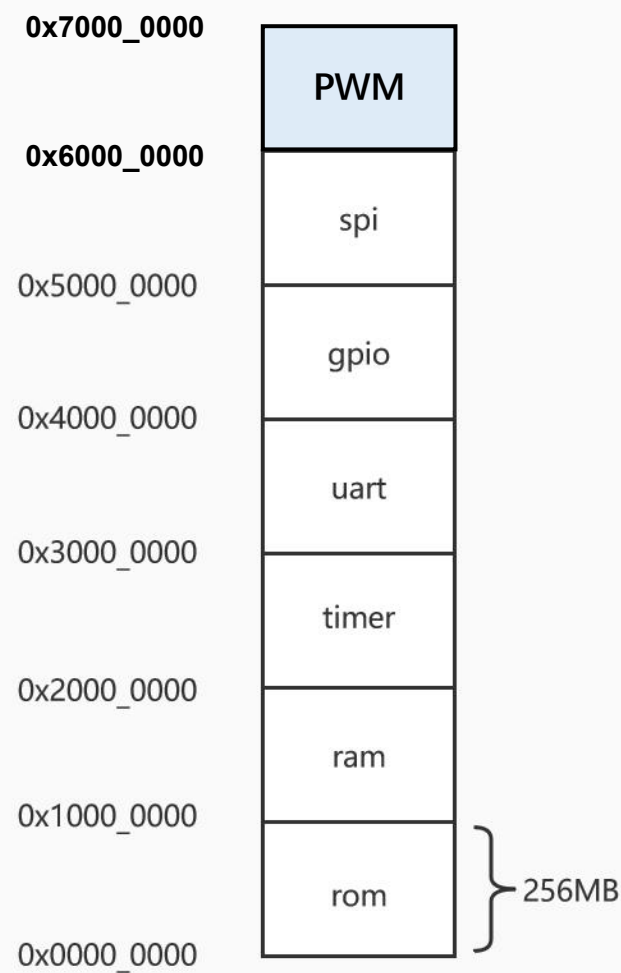
由于流片面积的限制，现需要将ROM与RAM移动到FPGA上：



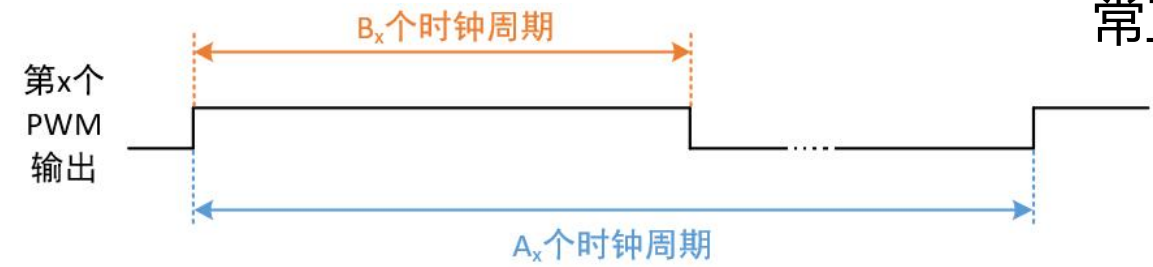
- 将原有的片内ROM和RAM删除，**禁止保留**片内 ROM 和RAM
- 新增总线到外部接口**桥接模块**，**支持处理器通过总线从外部ROM/RAM 读取指令与数据**，新增 8 位输入与 8 位输出外部接口用于芯片与 FPGA 交互，交互协议自定义。
- 实现 **FPGA 侧桥接模块**，与芯片桥接模块协议一致。
- 实现外部 ROM 模块，原有的 ROM 大小为 4096d x 32w，现需修改为 **256d x 32w**
- 实现外部 RAM 模块，原有的 RAM 大小为 4096d x 32w，现需修改为 **16d x 32w**

Tinyriscv修改2-PWM外设的添加

外设地址分配

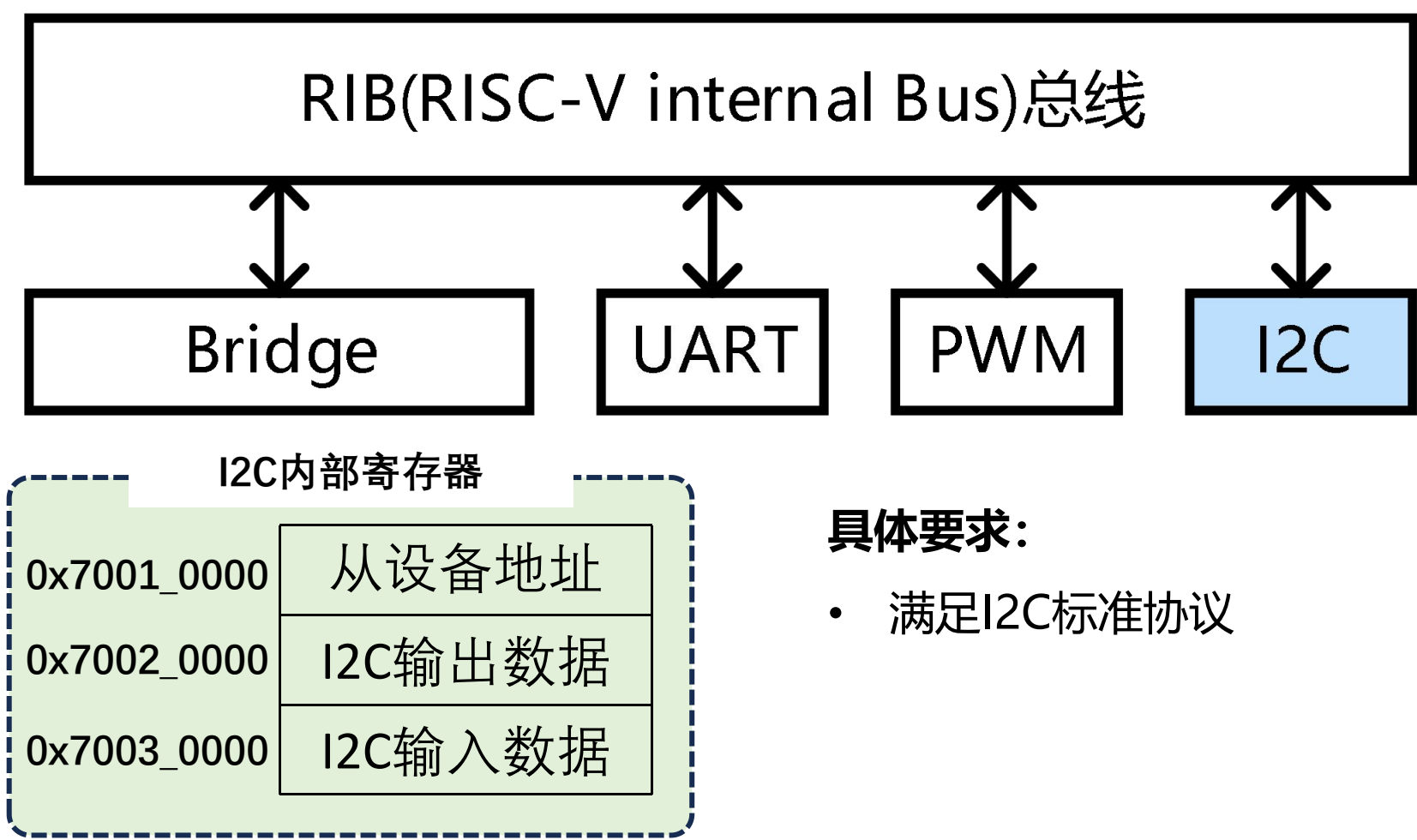
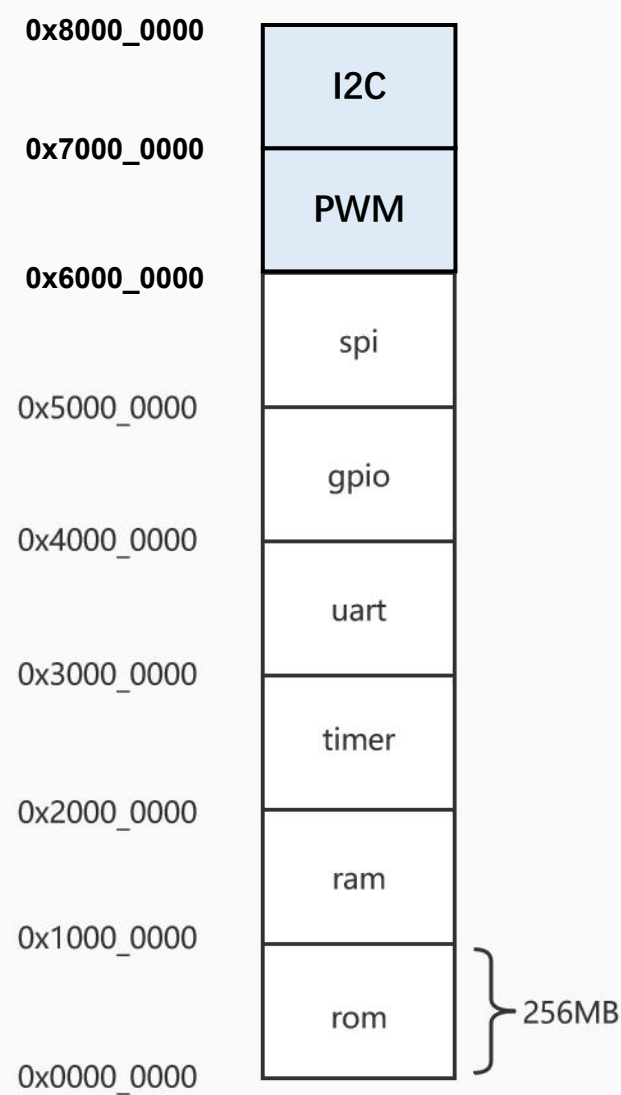


- 具体要求:
- 4个输出引脚
 - 可分别配置4个引脚的频率与占空比
 - C[3:0]使能4个输出引脚。例如C[0]为1时, 输出0正常工作, 否则保持为0



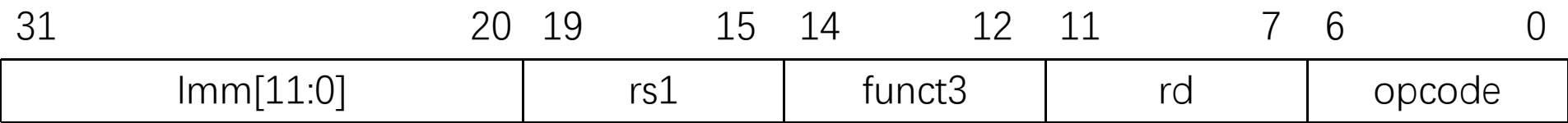
Tinyriscv修改3-I2C外设的添加

外设地址分配



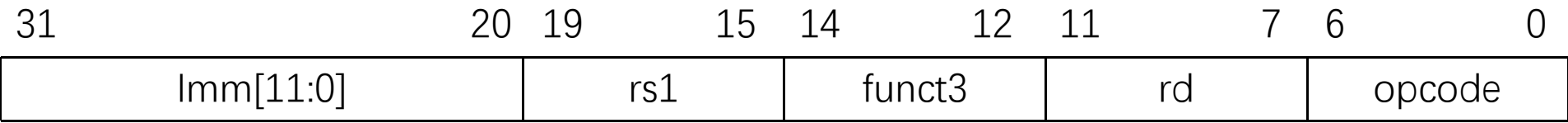
- 具体要求:**
- 满足I2C标准协议

Tinyriscv修改4-拓展指令添加0-Send ID



Inst	Opcode	funct3	功能描述
sID	0101111	0x0	接收到该指令时，芯片通过uart传回你的学号对应的ASCII码值。例如说学号为2022123456，则芯片依次回传 0x32 0x30 0x32 0x32 0x31 0x32 0x33 0x34 0x35 0x36

Tinyriscv修改5-拓展指令添加1-Read Temperature



Inst	Opcode	funct3	功能描述
rT	0101111	0x1	接收到该指令时，芯片通过I2C读取FPGA上的温度传感器芯片LM75的读数，并将温度读数存到x[rd] 中。

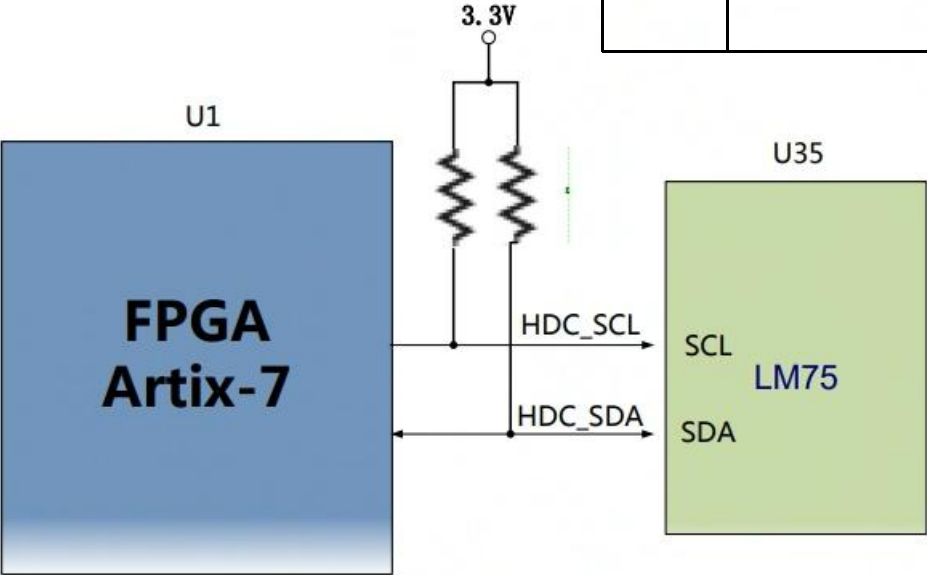


图 16-1 LM75 传感器原理图部分

表4. 温度数据输出格式

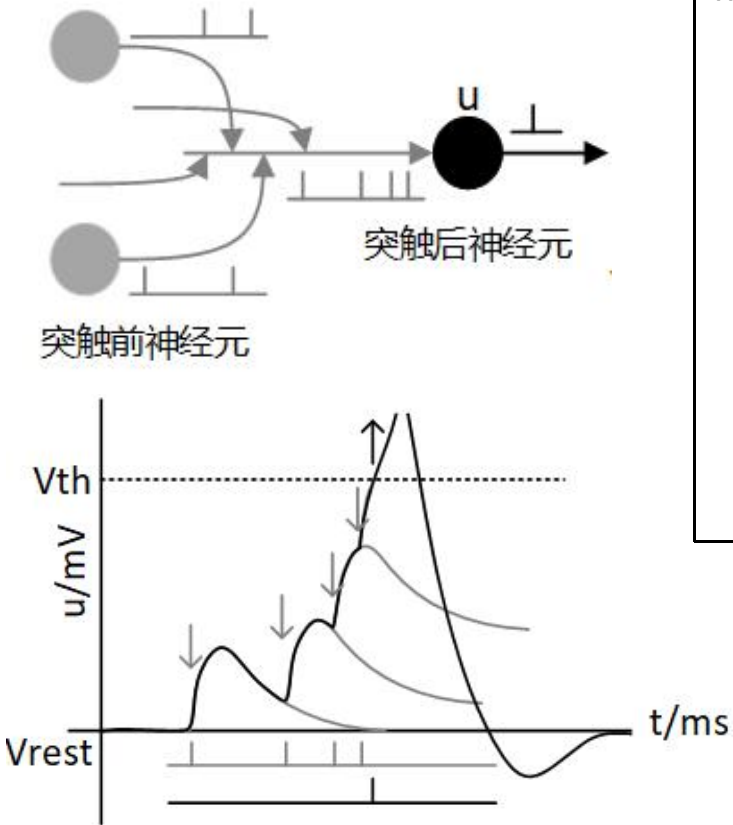
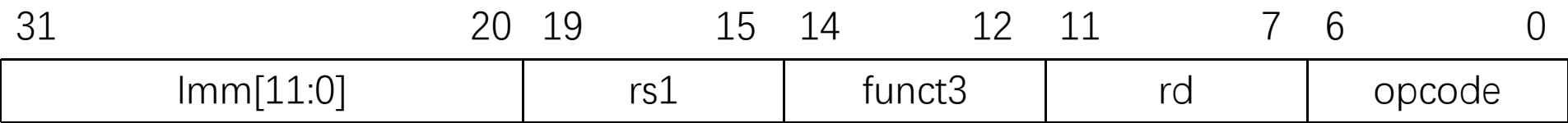
TEMPERATURE (°C)	DIGITAL OUTPUT	
	BINARY	HEX
+125	0111 1101 0XXX XXXX	7D0X
+25	0001 1001 0XXX XXXX	190X
+0.5	0000 0000 1XXX XXXX	008X
0	0000 0000 0XXX XXXX	000X
-0.5	1111 1111 1XXX XXXX	FF8X
-25	1110 0111 0XXX XXXX	E70X
-55	1100 1001 0XXX XXXX	C90X

X = 无关。

只需要存下并传回红框中8位读数即可

LM75的芯片手册在共享的清华云盘中

Tinyriscv修改6-拓展指令添加2-Integrated&Fire



Inst	Opcode	funct3	功能描述
if	0101111	0x2	<pre>if (Imm==0) { if (x[rs1] >= x31*) { 通过Uart将x[rs1][7:0]传回; x[rd] = 0; } else { x[rd] = x[rs1]; } } else { x[rd] = x[rs1] + sign_ext(imm[11:0]) }</pre>

* x31为riscv第32个通用寄存器

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- FPGA 基本使用

时间节点

- **FPGA验证时间到5月底**，会安排FPGA部分的考试，线下检查大家FPGA验证的成果
- **ASIC后端流程与仿真到7月底**，8月初会将GDS交予TSMC进行流片
- **ASIC芯片测试从11月到12月**

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FPGA功能验证

<https://cloud.tsinghua.edu.cn/d/8a852c44b6a2483a936e/>

1. **基本指令测试：**测试开源工程提供的指令测试程序，验证设计在FPGA上实现的功能正确性。（给大家提供了一些用于设计和调试用的程序，在`Baisc_Inst_Example`目录下）
2. **拓展指令测试：**测试添加的拓展指令的功能正确性。（`Extend_Inst_Example`）
3. **其它功能测试：**测试添加的模块功能正确性。（`Other_Example`）

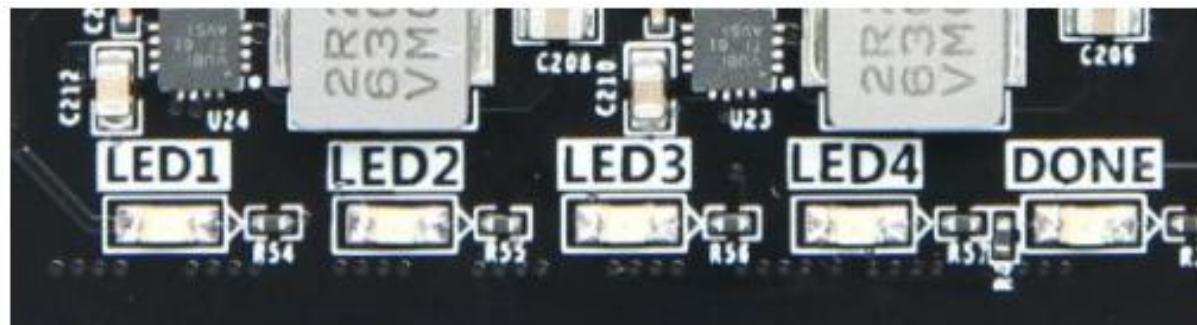


图 21-2 用户 LED 灯实物图

通过riscv的PWM模块控制LED灯

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Vivado下载和安装

<https://www.xilinx.com/support/download.html>

[Home](#) / [Adaptive Computing Support](#) / [Downloads](#)

Downloads

Licensing Help

NIC Software & Drivers

Vivado (HW Developer)

Vitis (SW Developer)

Vitis Embedded
Platforms

Alveo Packages

PetaLinux

Device Models

Documentation
Navigator

Version

[2022.2](#)

[2022.1](#)

[Vivado Archive](#)

[ISE Archive](#)

[CAE Vendor Libraries
Archive](#)

We strongly recommend using the latest releases available.

2021

2021.1

2021.2

2020

2020.3

2020.2

2020.1

2019

2019.2

2019.1

2018

Vivado下载和安装

<https://www.xilinx.com/support/download.html>

Vivado Design Suite - HLx Editions - 2019.1 Full Product Installation

Important

We strongly recommend to use the web installers as it reduces download time and saves significant disk space.

Please see [Installer Information](#) for details.

Note: Download verification is only supported with Google Chrome and Microsoft Edge web browsers.

Download Includes

Vivado Design Suite HLx Editions (All Editions)

Download Type

Full Product Installation

Last Updated

May 29, 2019

Answers

[2019.x - Vivado Known Issues](#)

Documentation

[Release Notes](#)
[What's New in Vivado](#)

Support Forums

[Installation and Licensing](#)

 [Vivado HLx 2019.1: WebPACK and Editions - Windows Self Extracting Web Installer \(EXE - 64.62 MB\)](#)

MD5 SUM Value : 743003070fb77857ad098bd6873bdf0b

 [Vivado HLx 2019.1: WebPACK and Editions - Linux Self Extracting Web Installer \(BIN - 115.05 MB\)](#)

MD5 SUM Value : 533000dc5324be422915eb4e93f9ce59

Download Verification

Digests

Signature

Public Key

 [Vivado HLx 2019.1: All OS installer Single-File Download \(TAR/GZIP - 21.39 GB\)](#)

Vivado下载和安装

Select Install Type



Please select install type and provide your Xilinx.com user ID and password for authentication.

User Authentication

Please provide your Xilinx user account credentials to download the required files.

If you don't have an account, [please create one](#). If you forgot your password, you can [reset it here](#).

User ID

Password

需要在xilinx官网注册一个账号

☒ Download and Install Now

Select your desired device and tool installation options and the installer will download and install just what is required. Downloaded installation files will be saved for future use. NOTE: Future installs using these downloaded files will be restricted to the options selected during this install. For access to all options later, choose "Download Full Image".

☐ Download Full Image (Install Separately)

The installer will download an image containing all devices and tool options for later installation. Use this option if you wish to install a full image on a network drive or allow different users maximum flexibility when installing.

Vivado下载和安装

Select Edition to Install



Select an edition to continue installation. You will be able to customize the content in the next page.

☒ Vivado HL WebPACK

Vivado HL WebPACK is the no cost, device limited version of Vivado HL Design Edition. Users can optionally add Model Composer and System Generator for DSP to this installation.

☐ Vivado HL Design Edition

Vivado HL Design Edition includes the full complement of Vivado Design Suite tools for design, including C-based design with Vivado High-Level Synthesis, implementation, verification and device programming. Complete device support, cable drivers and Documentation Navigator are included. Users can optionally add Model Composer to this installation.

☐ Vivado HL System Edition

Vivado HL System Edition is a superset of Vivado HL Design Edition with the addition of System Generator for DSP. Complete device support, cable drivers and Documentation Navigator are included. Users can optionally add Model Composer to this installation.

☐ Documentation Navigator (Standalone)

Xilinx Documentation Navigator (DocNav) provides access to Xilinx technical documentation both on the Web and on the Desktop. This is a standalone installation without Vivado Design Suite.

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< Back

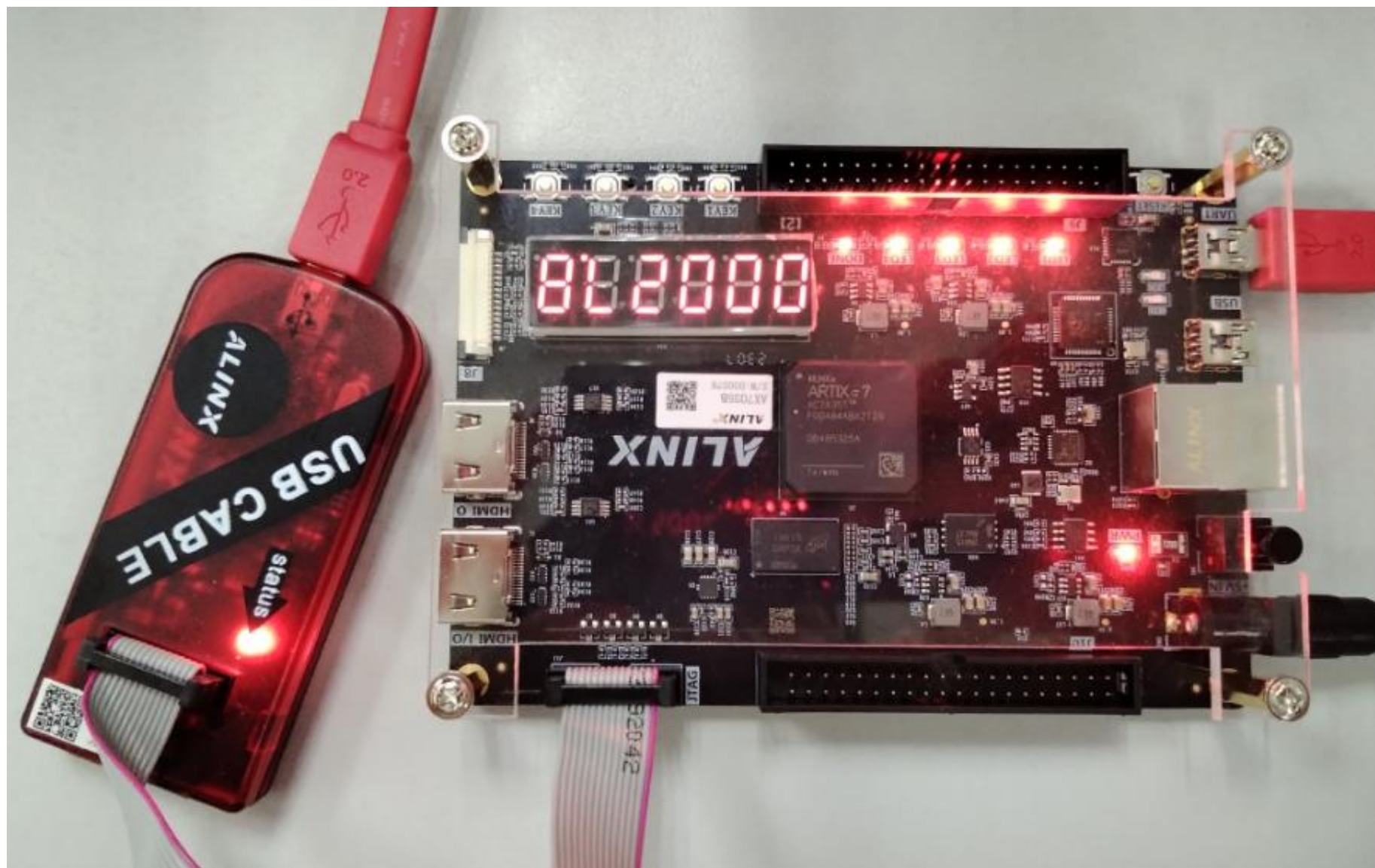
Next >

Cancel

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FPGA基本使用 - 开发板连接



FPGA基本使用 – 工程创建

Default Part

Choose a default Xilinx part or board for your project.



Parts | Boards

[Reset All Filters](#)

Category: Package: Temperature:
Family: Speed: Static power:

Search: (2 matches)

Part	I/O Pin Count	Available IOBs	LUT Elements	FlipFlops	Block RAMs	Ultra RAMs	DSPs	Gb Tran
xc7a35tfgg484-2	484	250	20800	41600	50	0	90	4
xc7a35tfgg484-2L	484	250	20800	41600	50	0	90	4



< Back

Next >

Finish

Cancel

- 创建工程时选择FPGA型号为xc7a35tfgg484-2

FPGA基本使用 – 引脚分配

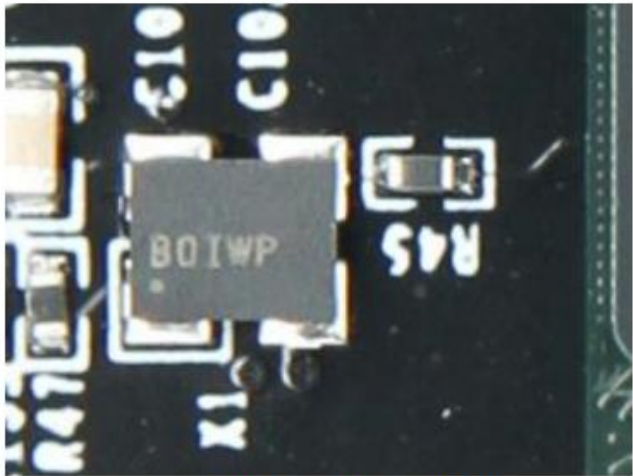
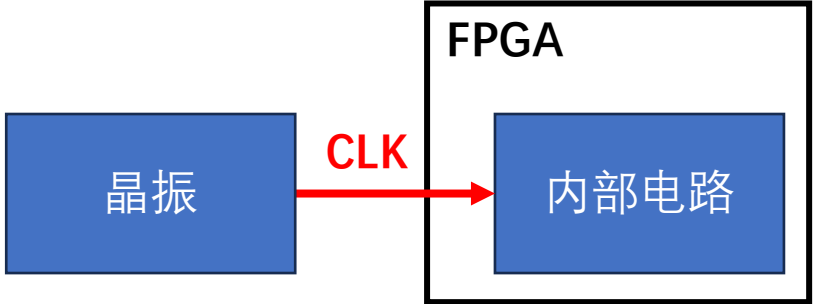
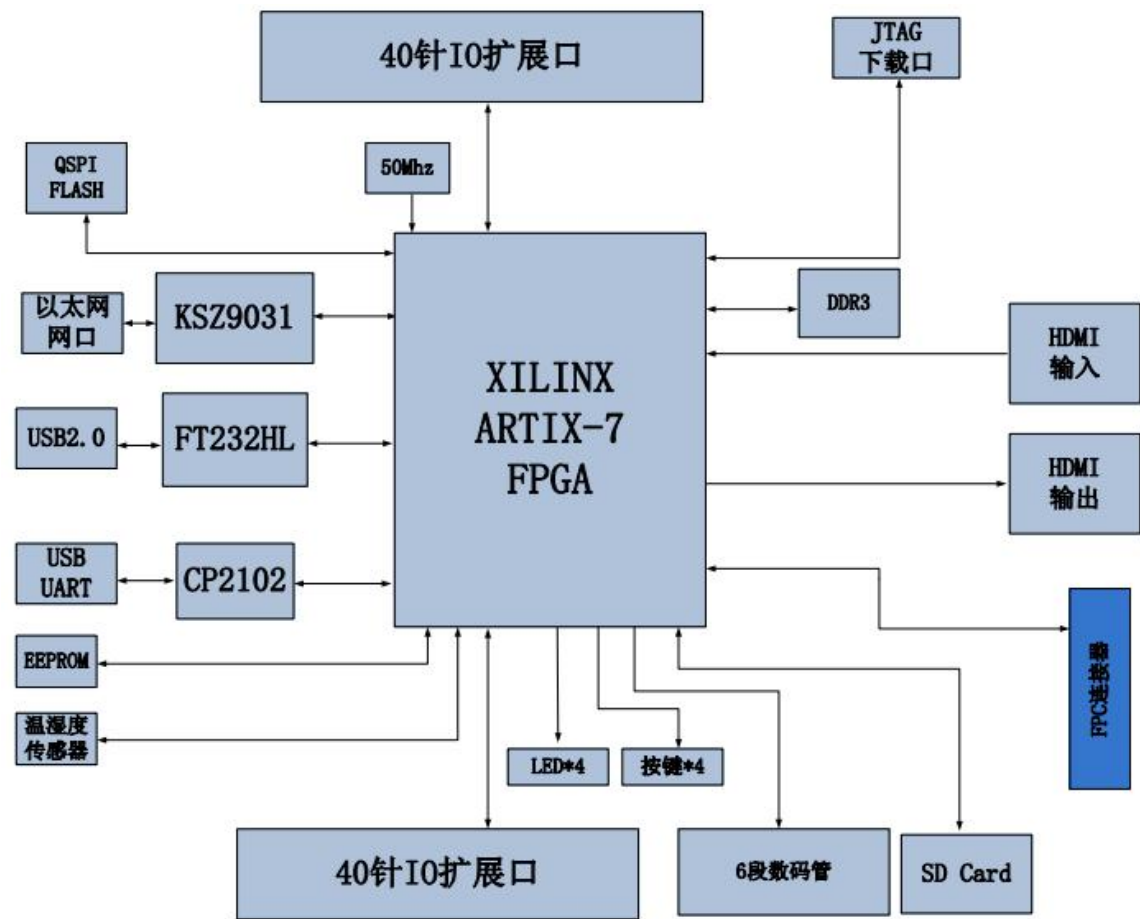


图 4.2 50M 有源晶振实物图

时钟引脚分配：

引脚名称	FPGA 引脚
FPGA_GCLK1	Y18

set_property IOSTANDARD LVCMOS33 [get_ports clk] ← 设置引脚电压

set_property PACKAGE_PIN Y18 [get_ports clk] ← 让clk连接到Y18引脚，接收来自晶振的时钟信号

FPGA基本使用 – 引脚分配

为了方便后续的FPGA功能验证检查，将部分需要用到的引脚分配统一一下

```
0
9 # 复位引脚
0 set_property IOSTANDARD LVCMOS33 [get_ports rst]
1 set_property PACKAGE_PIN M13 [get_ports rst]
2
3 # 程序执行成功指示引脚
4 set_property IOSTANDARD LVCMOS33 [get_ports succ]
5 set_property PACKAGE_PIN F19 [get_ports succ]
6
7 # 串口发送引脚
8 set_property IOSTANDARD LVCMOS33 [get_ports uart_tx_pin]
9 set_property PACKAGE_PIN G16 [get_ports uart_tx_pin]
0
1 # 串口接收引脚
2 set_property IOSTANDARD LVCMOS33 [get_ports uart_rx_pin]
3 set_property PACKAGE_PIN G15 [get_ports uart_rx_pin]
4
5 # PWM 引脚
6 set_property IOSTANDARD LVCMOS33 [get_ports PWM_o[0]]
7 set_property PACKAGE_PIN E21 [get_ports PWM_o[0]]
8
9 set_property IOSTANDARD LVCMOS33 [get_ports PWM_o[1]]
0 set_property PACKAGE_PIN D20 [get_ports PWM_o[1]]
1
2 set_property IOSTANDARD LVCMOS33 [get_ports PWM_o[2]]
3 set_property PACKAGE_PIN C20 [get_ports PWM_o[2]]
.
```

具体约束文件在：

<https://cloud.tsinghua.edu.cn/f/d8313c0a27774a818d98/>

FPGA 相关资源

链接: <https://pan.baidu.com/s/1eyfsU-5ID1B4Ob5LeYnsMw> 提取码: vtq0

☐	 01_demo_document.rar	2023-04-13 14:53	rar文件	1.14GB
☐	 02_学习教程之语言基础篇.rar	2023-04-13 14:53	rar文件	79.92MB
☐	 03_学习教程之参考篇.rar	2023-04-13 14:53	rar文件	24.97MB
☐	 04_原理图PCB结构图.rar	2023-04-13 14:53	rar文件	2.04MB
☐	 05_芯片手册.rar	2023-04-13 14:53	rar文件	60.52MB
☐	 06_模块资料.rar	2023-04-13 14:53	rar文件	25.31MB
☐	 07_软件工具及驱动.rar	2023-04-13 14:53	rar文件	365.39MB
☐	 08_其他资料.rar	2023-04-13 14:53	rar文件	57.62MB
☐	 09_microblaze.rar	2023-04-13 14:53	rar文件	11.33MB
☐	 11_4_hdmi_loop.rar	2023-04-13 14:53	rar文件	3.90MB
☐	 15_1_hdmi_char.rar	2023-04-13 14:53	rar文件	12.37MB
☐	 AX7035开发板检测指南V2.0.pdf	2023-04-13 14:53	pdf文件	1.90MB
☐	 AX7035开发板教程与程序对应关系说明.pdf	2023-04-13 14:53	pdf文件	303KB
☐	 AX7035开发板用户手册REV1.1.pdf	2023-04-13 14:53	pdf文件	1.90MB
☐	 ise_vivado_lic.zip	2023-04-13 14:53	zip文件	6KB

- 开发板用户手册
- 教程示例
- 没有FPGA基础的同学可以跟着教程熟悉基本操作

资源总结

- 开源RISC-V项目：tinyriscv (<https://gitee.com/liangkangnan/tinyriscv/tree/master>)
- 示例指令：<https://cloud.tsinghua.edu.cn/d/8a852c44b6a2483a936e/>
- FPGA资料：<https://pan.baidu.com/s/1eyfsU-5ID1B4Ob5LeYnsMw> 提取码：vtq0
- Xilinx Vivado下载链接：<https://www.xilinx.com/support/download.html>
- 串口驱动链接：<https://www.silabs.com/software-and-tools/usb-to-uart-bridge-vcp-drivers?tab=downloads> （下载第一个即可）